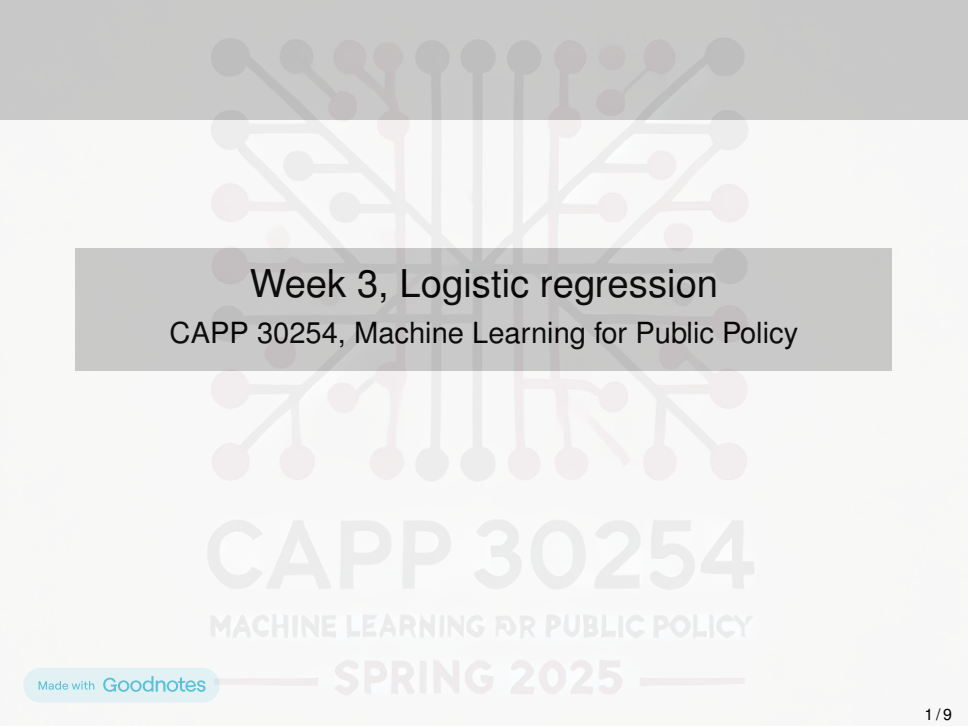


A faint, stylized network diagram in the background, consisting of numerous nodes (circles) connected by lines, forming a complex web-like structure. The nodes are in shades of gray, pink, and light blue.

## Week 3, Logistic regression

CAPP 30254, Machine Learning for Public Policy

CAPP 30254

MACHINE LEARNING FOR PUBLIC POLICY

— SPRING 2025 —

# Reminder: Linear models

Recall that linear regression models makes predictions using:

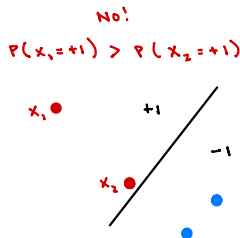
$$\hat{y} = \mathbf{w}^T \mathbf{x}$$

And linear classifiers make predictions:

$$\hat{y} = \begin{cases} +1 & \text{if } \mathbf{w}^T \mathbf{x} \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

given weight vectors  $\mathbf{w}$ .

Notice that is no probabilistic interpretation of these linear models.



# Probabilistic classification

Let  $\mathcal{D}$  be the set of training samples in a binary classification problem:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$$

where  $y_i = \{1, 0\}$ .

sample label

instead of  $\{+1, -1\}$

For a new sample  $\mathbf{x}$ , we estimate:

$$P(y = c_j | \mathbf{x}, \mathcal{D})$$

$$\hat{y} = \underset{c_j}{\operatorname{argmax}} P(y = c_j | \mathbf{x}, \mathcal{D})$$

$P(y = 1 \dots$   
 $P(y = 0 \dots$  → pick the biggest

for  $c_j = \{1, 0\}$  and assign to  $\mathbf{x}$  the label with the highest probability.

$$P(y = 1 | \mathbf{x}, \mathcal{D}) = 0.35 \text{ } \} \text{ binary}$$

$$\hat{y} = 0$$

$$P(y = 1 | \mathbf{x}, \mathcal{D})$$

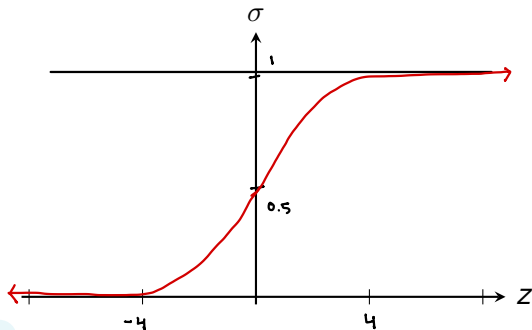
estimate  
the label

given the sample  
and training data

# Logistic/sigmoid function

The probability that  $\mathbf{x}$  belongs to a class is computed using the *logistic* or *sigmoid* function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



$\sigma: \mathbb{R} \rightarrow (0, 1)$   
We can use  $\sigma$  to  
model probabilities

# Computing probabilities

The probability that  $\mathbf{x}$  belongs to a class is computed using the *logistic* or *sigmoid function*:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The probability that  $y = 1$  is computed as:

Pick  $y$  then compute  $P(y=1 | \mathbf{x}, \mathcal{D})$

$$P(y = 1 | \mathbf{x}, \mathcal{D}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$
$$= \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$$

$\exp(z) = e^z$

# Computing probabilities

The probability that  $\mathbf{x}$  belongs to a class is computed using the *logistic* or *sigmoid function*:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The probability that  $y = 1$  is computed as:

$$P(y = 1 \mid \mathbf{x}, \mathcal{D}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

What is the probability that  $P(y = 0 \mid \mathbf{x}, \mathcal{D})$ ?

$$P(y = 0 \mid \mathbf{x}, \mathcal{D}) = 1 - \sigma(\mathbf{w}^\top \mathbf{x}) \quad \text{binary classification}$$

# How do we learn the weights $\mathbf{w}$ ?

Just like before, we minimize the loss function to find the weight vector  $\mathbf{w}$ . In logistic regression, the loss function is also called the *cross-entropy loss* or the *log loss*.

$$\ell(\mathbf{w}) = - \sum_{i=1}^n \log P(y_i | x_i, \mathbf{w})$$

Then the best weight vector  $\mathbf{w}$  is:

$$\mathbf{w}_{LR} = \operatorname{argmin}_{\mathbf{w}} \ell(\mathbf{w})$$

No close-form solution  
→ gradient descent

We can also view this problem as maximizing the likelihood of observing the training data.

$\operatorname{argmax}$  LL: max agreement between model and data

$\operatorname{argmin}$  LR: min disagreement between model and data (min  $\ell$ )

# Multiclass logistic regression

Logistic regression can be extended to a multiclass classification problem.

Let  $\mathcal{D}$  be the set of training samples in a classification problem:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$$

where  $y_i = \{c_1, c_2, \dots, c_k\}$ .  
*0, 1, 2, ..., k-1*

For a new sample  $\mathbf{x}$ , we estimate:

$$P(y = c_j \mid \mathbf{x}, \mathcal{D})$$

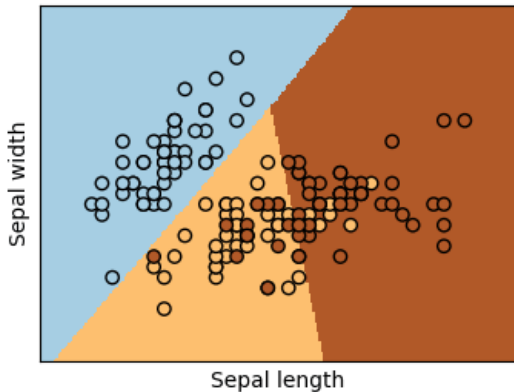
*same formula as before*

for  $j = 1, \dots, k$  and assign to  $\mathbf{x}$  the label with the highest probability.



# Multiclass logistic regression

Fisher's iris flower dataset



1

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<sup>1</sup>Logistic classification example with Fisher's iris flower dataset implemented in scikit-learn.

What's next?

X, vocabulary

- Fundamentals
- supervised classification
  - least squares
  - Perceptron
  - regularization (ridge regression)
  - polynomial boundaries
  - logistic regression
- Related topics
  - gradient descent



Model evaluation and selection

- bias - variance trade off

\* NEXT 3 LECTURES

How to pick a model

Pick parameters like  $\eta$

Evaluate a model

2

<sup>2</sup>Generated by GPT-4, DALL-E 3 after the prompt: "Hi, could you please make a minimalistic logo for a machine learning class..."